INSA INSTITUT NATIONAL DES SCIENCES APPLIQUÉES LYON

Domaine Scientifique de la DOUA
20, Avenue Albert Einstein - 69100 Villeurbanne

Business Intelligence
Business Intelligence

IDENTIFICATION

CODE :	GI-4-S1-EC-BIN
ECTS :	2

HOURS

Cours :	0h
TD :	4h
TP :	14h
Projet :	0h
Evaluation :	2h
Face à face pédagogique :	20h
Travail personnel :	12h
Total :	32h

ASSESMENT METHOD

Written exam at the end of the tutorial part ("TD")
Project evaluation : collective presentation and written report

TEACHING AIDS

Slides

TEACHING LANGUAGE

English

CONTACT

M. DUMITRESCU Emil :
emil.dumitrescu@insa-lyon.fr

Mme SERNA Audrey :
audrey.serna@insa-lyon.fr

AIMS

GI-4-S1-UE-GEDE
GI-4-S1-EC-BIN
General engineering skills:
- Develop an experimental approach;
- Identify, formulate and solve a complex engineering problem.

Specific Industrial Engineering skills:
- Observe, measure, analyze and interpret an activity or system from data
- Modeling, designing a system of information, decision-making and production of goods and services;
- Evaluate, prototype or simulate a system.

Abilities :
Collect customer needs for analysis; Identify sources of production data useful for an analytical approach; Process voluminous information for analysis; Exploit the data in the perspective of the analyzed activity: Carry out a project, working in a group.
Knowledge: operational and analytical database schemas; Elementary actions and sequencing for data transformation; Methods and tools for visualizing information from data.

CONTENT

DATA WAREHOUSE
1. Dimensional modeling for decision support systems
2. The Extract-Transform-Load (ETL) task : data cleaning, fusion and restructuration methods, ETL tools (Tableau)
3. Query tools and methods : reporting, scorecards, etc.

BIBLIOGRAPHY

Systèmes d'information décisionnels, E. Ferragu, 2013 (available at Doc'INSA)
- Piloter l'entreprise grâce au data warehouse, Jean-Michel Franco, Sandrine de Lignerolles , 2000
(available at the Industrial Eng. dept. local library)

PRE-REQUISITES

Relational Databases, SQL